

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 3 teacher notes

Sequencing note

Topic 3.1 must come first: variables make sense only as names for quantities a pattern already varies. Topic 3.2 adds the claim (an equation) and the range (an inequality). Topic 3.3 closes the loop on real rate cards. Across the three weeks, insist on the vocabulary table from Topic 3.1 - "variable", "coefficient", "term", "expression", "equation" - because half the errors in this unit are naming errors.

Teaching strategies from the guide

- Use the guide's brainstorming mode for the matchstick pattern: let the class find the growth step before you name it.
- Run the fare desk (Topic 3.2) as guided practice: the tutor circulates while pairs trade equations, and checks the check, not just the answer.
- Use question-answer on the translation table (Topic 3.3): cover the algebra column and have the class rebuild it from the everyday phrases.

Common misconceptions to probe

1. **"A letter is a thing to solve for."** Write $2t + 1$ on the board and ask what t is. The productive answer is "whatever number of triangles you have". If the class stalls, feed $t = 1, 2, 3$ and watch the machine work.
2. **"The equals sign means the answer comes next."** The chain $60 + 15k = 195 = 135 = 9$ appears on scripts every year. Draw the balance; ask whether 195 equals 135.

3. **"Multiplying by a negative follows the same rules."** Give $x = 5$ and ask for $-2x$; most will say -10 is "less than" something vague. Put 5 and -10 on the number line against the original inequality and let the flip show itself.
4. **"Algebra is separate from real maths."** Bring two real rate cards to the last session and let the class discover that one comparison settles every quantity at once.

Running the activities

- **Rule detective (Topic 3.1):** the swap-and-check step (testing another pair's rule on stages 5 and 6) is the point; do not let pairs skip it.
- **The fare desk (Topic 3.2):** rotate after exactly three problems so every pair both writes and solves.
- **Rate card clinic (Topic 3.3):** insist the cards come from real shops or services the trainees know; invented round numbers hide the translation difficulty.

Practical work

Group work in the three activities; a group assignment building the class's rate-card wall (each card with its expression pinned beside it); the individual assignment is the Topic 3.3 practicum collection; presentations come from the clinic defences.

Assessment guidance

The bank's traps are deliberate: MCQ-07 and RRQ-6 test the sign flip, RRQ-7 the equals-sign chain, and ERQ-3's break-even is the unit's central transfer. When marking ERQ-2, accept no answer without the look-back check: the whole-number rounding (100 pupils, exactly 20,000 rupees) is where arithmetic meets the budget.

Practicum links

Trainees collect three artefacts this unit: a growing pattern from the school (Topic 3.1), a posted rate (Topic 3.2), and a full rate card with a flat-rate comparison (Topic 3.3). ERQ-5 consumes the third; insist it is real, dated and specific.

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